Alejo García-Sassi

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SUMMARY

- **Broad interest in mathematics**: dynamical systems, geometry and topology, ergodic theory; and their application to other fields (i.e. astrodynamics).
- **1 single-authored published article**.
- **2 long-form research articles** under consideration by high-quality journals.
- **Co-supervised 1 Master's student** (completed February 2026).
- **Solid speaking activity:** Invited talks, seminars, and a minicourse at international venues.
- **13 years of teaching experience**, 11 at university level; consistently strong evaluations.
- **Strong outreach profile**: 5 years with TMU project (workshops, lectures, science events), science-communication columns, interviews, and a popular-science book in preparation.
- **Robust fellowship record**, including Master's, PhD, and postdoctoral awards.
- **3x visiting PhD at IMJ-PRG - Sorbonne Université**.
- **Ongoing collaboration** with researchers at **IMJ-PRG**, **USP** and **UdelaR**.
- Native **Spanish** speaker, fluent in **English and Portuguese**, competent in **French**.

POSITIONS

POSTDOCTORAL RESEARCHER | *FAPESP – São Paulo, BR* Oct25 - **Current**
Competitive call: position granted among over 20 applicants worldwide.
Project: *Dynamics and geometry in low dimensions*.

JUNIOR PROFESSOR | *IMERL – Montevideo, UY* Aug19 – **Current** (on leave)
Competition-awarded permanent position since 2023.

POSTDOCTORAL RESEARCHER | *PEDECIBA – Montevideo, UY* Oct24 - Sep25
Ranked 1st in the selection process.
Project: *Rotation sets and actions on curves for higher genus surface homeomorphisms*.

JUNIOR PROFESSOR | *CMAT – Montevideo, UY* Aug23 – Sep24
Junior professorship awarded during the transition from my PhD to PostDoc status.

TEACHING ASSISTANT | *CMAT – Montevideo, UY* Aug16 – Mar18
Teaching duties during my bachelor's training (resigned to accept master's scholarship).

TEACHING ASSISTANT | *IMERL – Montevideo, UY* Mar15 – Jul19
Teaching duties during my bachelor's and Master's training.

MATH OLYMPIAD INSTRUCTOR | *Los Pinos – Montevideo, UY* Feb14 – Feb18
Training the best national students for the International Mathematical Olympiad.

MATH OLYMPIAD INSTRUCTOR | *OPM – Ciudad de Panamá, PA* Jan14 – Feb14
Training the National Panamanian Team for the International Mathematical Olympiad.

Head of Math Olympiad Training Program | *ELEF – Montevideo, UY* Mar13 – Dec17
Mentored an eventual Math Olympiad National Champion, and several national medalists.

EDUCATION

PhD in applied Mathematics | *USP-UdelaR – São Paulo, BR* 2024
- Graduated with honors.
- Visiting PhD at Sorbonne Université, working with Pierre-Antoine Guihéneuf.
- Thesis: *Fully chaotic conservative models for some torus homeomorphisms.*
- Advisors: Alejandro Passeggi – Fábio Armando Tal.

MSc In Mathematics | *UdelaR – Montevideo, UY* 2020
- Graduated with honors, due to original results.
- Thesis: *Fixed points for branched covering maps of the plane.*
- Advisor: Juliana Xavier.

Bachelor in Mathematics | *UdelaR – Montevideo, UY* 2017
- Monograph: *Homeomorfismos del plano: toda órbita periódica enlaza un punto fijo.*
- Advisor: Juliana Xavier.

PUBLICATIONS AND PREPRINTS

Fully chaotic conservative models for some torus homeomorphisms 2024
With Fábio Armando Tal | *arXiv:2404.02341*
Under consideration by **Memoirs AMS**

Geodesic tracking and the shape of ergodic rotation sets 2023
With Pablo Lessa and Pierre-Antoine Guihéneuf | *arXiv:2312.06249*
Under Consideration by **ASENS**

Fixed Points for branched covering maps of the plane 2021
Published in **Fundamenta Mathematicae** 254 (2021), 1-14

SUPERVISION AND PROFESSIONAL TRAINING

Lara de Assumpção Maffei Pierobon 2025-2026
Master student | *Topological horseshoes*
Co-supervision with Fábio Armando Tal. Defended in 2026.

TMU project 2017 – 2021
Including workshops for high-school teachers on math problem-solving.

INVITED TALKS

- *Chaotic models for hyperbolically acting homeos of higher genus surfaces* Sch. 2026
  Surf. Dyn. and Fine Curve Graphs Workshop | **Sorbonne Université** – Paris, France
- *Where is the caos? Models for surface dynamics* Sch. 2026
  II meeting of Postdoctoral Researchers | **USP** – São Paulo, Brazil
- *Chaotic models for higher genus surface homeomorphisms* 2026
  Seminário de Sistemas Dinâmicos | **USP** – São Paulo, Brazil

- *Where is the caos? Models for surface dynamics* 2025
  9° Coloquio Uruguayo de Matemática | **UdelaR** – Montevideo, Uruguay
- *Fully chaotic models for higher genus surface homeomorphisms* 2025
  8° Escola Brasileira de Sistemas Dinâmicos | **IMPA** – Rio de Janeiro, Brazil
- *Explicit uniform bounds for fixed elliptic islands* 2025
  Intl. Conf. on Advanced Topics in Dynamical Systems | **Nankai University** – Tianjin, China
- *Fully chaotic models for some torus homeomorphisms* 2024
  Seminário de Sistemas Dinâmicos | **IMPA** – Rio de Janeiro, Brazil
- *Fully chaotic conservative models for some torus homeomorphisms* 2024
  Seminario de Sistemas Dinámicos | **UdelaR** – Montevideo, Uruguay
- *Uniform bounds in strictly toral dynamics* 2023
  Surfaces in Banyuls | **Observatoire Océanologique de Banyuls-sur-mer** – France
- *A little bit of forcing in strictly toral dynamics* 2023
  Séminaire de Dynamique Sauvage | **Sorbonne Université** — Paris, France
- *Applications of Haefliger's Theorem* 2022
  School: Foliaciones y flujos de Anosov en dimensión 3 | **UdelaR** – Montevideo, Uruguay
- *Topologically chaotic surface homeomorphisms* 2021
  Encuentro de Estudiantes de Posgrado | **UdelaR** – Montevideo, Uruguay
- ***Minicourse** - Topics in plane topological dynamics* (with Juliana Xavier) 2019
  7° Coloquio Uruguayo de Matemática | **UdelaR** – Montevideo, Uruguay
- *Fixed point theorems for branched covering maps of surfaces* 2019
  7° Coloquio Uruguayo de Matemática | **UdelaR** – Montevideo, Uruguay
- *Growth rate of periodic orbits for branched covering maps* 2019
  Seminario de Sistemas Dinámicos | **Universidad de Santiago de Chile** – Chile
- *Combinatorial invariants* 2017
  6° Coloquio Uruguayo de Matemática | **UdelaR** – Montevideo, Uruguay
- *Linking fixed points in plane homeomorphisms* 2017
  Seminario de Sistemas Dinámicos | **UdelaR** – Montevideo, Uruguay

TEACHING

Graduate courses

- *Chaotic Models and Rotation Theory for Surface Homeomorphisms* Sched. 2026
  Short course – Lecturer. | **USP**
- *Endomorphism dynamics* 2025
  Seminar. Assisting Juliana Xavier | **IMERL**
- *Applications of Linear Algebra* 2024
  Coordination + problem class instructor | **IMERL**
- *Topics in Topological Plane Dynamics* 2019
  Conference minicourse. Co-lecturer | **IMERL**

Undergraduate courses

Course Coordination

- *Discrete Mathematics* | **CMAT** 2024
- *Complex Analysis* | **CMAT** 2024
- *Probability and Statistics* | **IMERL** 2023
- *Multivariable Calculus* | **IMERL** 2021
- Introduction to Topology | **CMAT** 2017
- Geometry of Curves and Surfaces | **CMAT** 2017

Lecturer

- Vector Calculus | **IMERL** 2025
- Geometry and Linear Algebra | **IMERL** 2024
- Multivariable Calculus | **IMERL** 2021, 2022

Problem Class Instructor

- Geometry and Linear Algebra | **IMERL** 2015, 2025
- Complex Analysis | **IMERL, CMAT** 2018, 2024
- Discrete Mathematics | **IMERL, CMAT** 2015, 2017, 2018, 2024
- Differential Equations | **IMERL** 2019, 2020, 2021, 2024
- Probability and Statistics | **IMERL** 2023
- Vector Calculus | **IMERL** 2020
- Introduction to topology | **CMAT** 2017
- Mathematics for Biologists | **CMAT** 2016
- Single-variable Calculus | **IMERL** 2016, 2017, 2019
- Multivariable Calculus | **IMERL** 2016

Institutional evaluations

Consistently strong student evaluations, averaging above **4.72/5 over 165+ evaluations** across multiple courses in the last five years – Verifiable at IMERL - UdelaR.

EDUCATIONAL RESOURCES

Revista TMU 2020

Compilation of notes prepared specifically for TMU workshops

Tópicos en dinámica topológica plana 2019

Conference minicourse notes | *7° Coloquio Uruguayo de Matemática*
With Juliana Xavier.

Seminarios de problemas 2017 – 2021

Over 30 notes, each containing a brief theoretical overview and a curated set of problems.

Talleres - Campamento matemático 2014 - 2018

Math Summer Camp | *Centro Educativo Los Pinos*

OUTREACH

Outreach project 2017 – 2021

Coordination, organization and teaching in *Talleres Matemática UdelaR* **TMU.**

- Problem-solving workshops for IMO preparation (students and teachers).
- Organization of plenary talks and annual science recreational days, receiving in total, thousands of students and teachers from across the country.

Science popularization

- **Science-communication columns** in *Núcleo Delirante #3, #6 - Desembarco TV.* . 2025
- **Interview** #12 of the series *Mis profes en Fing.* 2024
- **Contribution to exhibitions**: *Íngeniería de Muestra* 2022, 2024
  Día internacional de la Matemática – Fing 2024
  Imaginary – Un viaje por la matemática , 2015
- **Outreach seminar**: Inter-School Mathematical seminar - The British Schools. 2015

Event organization

Mathematical Olympiad National Final | *Facultad de Ingeniería* 2019

Community service

Jury member and tutor | *Uruguayan Math Delegation* 2011-2015
In national and international mathematical olympiads.

Popular-science book

A collection of short stories built around rich mathematical concepts. In preparation

Fellowships and research internships

I have been awarded more than 160K USD in graduate and postdoc fellowships, and more than 25K USD in previous scholarships.

Fellowships

- **FAPESP** postdoc fellowship | *332K BRL ∼ 61K USD* Oct25 – Current
- **PEDECIBA** postdoc fellowship | *934K UYU ∼ 23K USD* Oct24 – Sep25
- **CAP** PhD fellowship | *2.38M UYU ∼ 59K USD* Apr20 – Mar24
- **CAP** Master's fellowship | *640K UYU ∼ 19K USD* Mar18 – Feb20
- **IMPA** Summer School scholarship | *2400BRL ∼ 800 USD* Jan17 – Feb17
- **ELEF** *Beca a la excelencia académica* scholarship | *504K UYU ∼ 25K USD* Mar10 – Dec12

Research internships

- **IMPA** | *with Alejandro Kocsard* Aug24
- **IMJ-PRG, Sorbonne Université** | *with Pierre-Antoine Guihéneuf* Jun24
- **IMJ-PRG, Sorbonne Université** | *with Pierre-Antoine Guihéneuf* Sep23 – Dec23
- **IMJ-PRG, Sorbonne Université** | *with Pierre-Antoine Guihéneuf* Sep22 – Oct22
- **USP** | *with Fábio Armando Tal* Feb22 – Aug22
- **USACH** | *with Andrés Navas* Aug19

Honours and awards

- **2x Honourable Mentions:** International Mathematical Olympiad 2011, 2012
- **3x Bronze Medals:** Iberoamerican Mathematical Olympiad 2012
  Iberoamerican Physics Olympiad 2012
  Asian-Pacific Mathematical Olympiad 2012
- **2x Gold Medals**: National Mathematical Olympiad 2011, 2012

Languages

Spanish | *Mother tongue*
English | *Proficient*
Portuguese | *Proficient*
French | *Competent*